

Where the wedge Gibbs–Thomson deficit comes from

A phase-field calibration audit of `lunar_regolith_DSM`

Working note — batch 2026-08-07__07.26.38_wedge_bc

2026-08-25

1. The measurement

The wedge geometry places an apex-centred ice annulus in a tapering pore, with the virtual apex at $y = L_y/2$. On the centreline the interface normal is therefore *exactly* $\pm\hat{x}$, so v_n reduces to a one-dimensional displacement rate. `postprocess/wedge_gt_velocity.py` measures it and compares against

$$v_n = \frac{\sigma \rho_{vs}(T)/\rho_{vs}(T_0) - d_{0,\text{sub0}} \chi}{\beta_{\text{sub0}}}, \quad \sigma = \frac{\rho_v - \rho_{vs}(T)}{\rho_{vs}(T)}. \quad (1)$$

Across all nine runs the solver recovers 0.826 (inner) / 0.811 (outer) of that rate, flat over 90 days and flat across a residual driving force $\sigma - d_0\chi$ spanning two orders of magnitude.

Free two-parameter fit. Regressing $\sigma = \beta v_n + d_0\chi$ over all nine runs $\times$ both fronts $\times$ 1000 snapshots ($n = 3492$; condition number 2.6, because the two fronts carry opposite-sign χ):

	fitted	vs. requested
d_0	$1.0171 \times 10^{-9} \text{ m}$	1.0005 $\times$ -d0_sub0
β	$7.227 \times 10^5 \text{ s/m}$	1.2205 $\times$ -beta_sub0

The curvature physics is exact to 0.05 %. The entire deficit lives in β .

2. β lands on the *bare* Karma–Plapp coefficient

The phase equation, after dividing the residual by $3 \text{ mob}/\epsilon$, is

$$\tau \partial_t \phi = \epsilon^2 \nabla^2 \phi - f'(\phi) + \lambda g(\phi) u, \quad u = \frac{\rho_v - \rho_{vs}(T)}{\rho_{\text{ice}}}. \quad (2)$$

Its sharp-interface limit is $u_{\text{int}} = d_0\chi + \beta_{\text{real}}V$ with

$$d_0 = \frac{a_1 \epsilon}{\lambda}, \quad \beta_{\text{real}} = \underbrace{\frac{a_1 \tau}{\lambda \epsilon}}_{\beta_{\text{bare}}} - C_{\text{corr}}. \quad (3)$$

C_{corr} is an $O(\epsilon)$ term generated by u varying *inside* the diffuse interface. `lunar_main.c:829` pre-inflates τ by precisely what it assumes C_{corr} to be, so that the two cancel and $\beta_{\text{real}} = \beta_{\text{sub0}}$:

$$\tau_{\text{sub}} = \epsilon \lambda \left(\underbrace{\frac{\beta_{\text{sub}}}{a_1}}_{82.2\%} + \underbrace{\frac{a_2 \epsilon}{\text{diff_sub}}}_{13.1\%} + \underbrace{\frac{a_2 \epsilon}{\text{dif_vap}}}_{4.7\%} \right). \quad (4)$$

Measured: $\beta_{\text{real}} = \beta_{\text{bare}}$ to 0.3 %, i.e. $C_{\text{corr}} \approx 0$. **Nothing is subtracted, so the inflation survives**, and $\beta_{\text{bare}}/\beta_{\text{sub0}} = 1.2165 \Rightarrow$ predicted ratio 0.8220.

This is a defect in a *model parameter definition*, not in the prediction. The prediction is the target physics and it is right; the solver simply does not realise the β it was asked for.

Grid pinning is ruled out independently: the deficit is a constant *factor*, identical on fronts whose speeds differ $60\times$. A friction or threshold cannot do that.

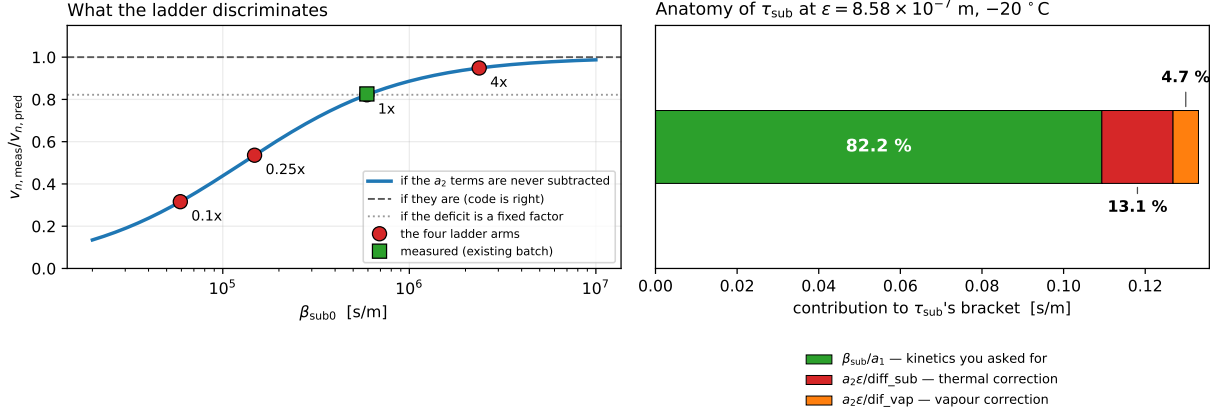


Figure 1: **Left:** the three hypotheses are separated by how the ratio varies with $\beta_{\text{sub}0}$, because the a_2 inflation is additive while $\beta_{\text{sub}0}$ is not. **Right:** the anatomy of Eq. (4). The thermal term is the larger of the two corrections and is the one that is wrong on its face (§4).

3. ξ_v does *not* enter the interface condition

In the inner region, moving frame, the vapour equation reads

$$\underbrace{\phi_a \frac{V}{W} \partial_\eta \rho_v}_{\text{storage}} = \underbrace{\frac{\xi_v}{W^2} \partial_\eta (D_v \phi_a \partial_\eta \rho_v)}_{\text{diffusion}} + \underbrace{\xi_v \rho_{\text{ice}} \frac{V}{W} \partial_\eta \phi}_{\text{ice source}}.$$

ξ_v multiplies diffusion *and* source identically, so it cancels between them and survives only in storage/diffusion, which here is $VW/(\xi_v D_v) = 2.4 \times 10^{-11}$. **The unscaled dif_vap in Eq. (4) is correct.** Substituting $\xi_v \text{dif_vap}$ would be wrong on principle and catastrophic in practice:

τ_{sub} bracket	$\beta_{\text{bare}}/\beta_0$	predicted ratio
as coded	1.2165	0.8220
with $a_2\epsilon/(\xi_v \text{dif_vap})$	58.13	0.0172

ξ_v would only reach the interface condition once $V \gtrsim \xi_v D_v/\epsilon = 2.5 \times 10^{-2}$ m/s — about 4×10^{10} times the actual velocities.

3.1 What ξ_v *does* cost: mass conservation

The coded vapour equation conserves $M_{\text{mod}} = \int \phi_a \rho_v + \xi_v \rho_{\text{ice}} \int \phi$, not the physical $M_{\text{phys}} = \int \phi_a \rho_v + \rho_{\text{ice}} \int \phi$. Equivalently, dividing through by ξ_v : the ice source and the diffusion are physical and only the vapour *storage* term is inflated by $1/\xi_v$. So mass is exact in the quasi-static limit, which is the regime the model is designed for. Magnitude:

	-20°C	0°C
$M_{\text{vap}}/M_{\text{ice}}$	1.9×10^{-6}	5.5×10^{-6}
worst-case error, $(1/\xi_v)M_{\text{vap}}/M_{\text{ice}}$	1.9×10^{-3}	5.5×10^{-3}
observed over 90 d (three runs)	$1.4\text{--}4.8 \times 10^{-5}$	

So: acknowledge and document, do not chase. It is two to three orders of magnitude below the effect under investigation. If exact conservation is ever needed, the only real fix is $\xi_v = 1$ with a stiffer vapour solve — ξ_v is a stiffness knob ($D_v/dx^2 = 5.2 \times 10^7 \text{ s}^{-1}$ unscaled), and its price is a wrong vapour *transient* plus this bounded bookkeeping error.

4. The thermal a_2 term is wrong on its face

`a2*eps/diff_sub` treats latent heat as if it drove the phase field at unit strength. It does not: temperature reaches u only through $\rho_{vs}(T)$, so that channel carries a Clausius–Clapeyron factor. Working the inner balance for each channel and reading off $du/d(VW)$:

channel	correct coefficient	value
vapour	$1/D_v$	5.27×10^4
	<i>coded:</i> <code>1/dif_vap</code>	4.59×10^4 ✓
thermal	$(d\rho_{vs}/dT) L_{\text{sub}}/k$	$1.0 \times 10^2 (k_{\text{ice}}) - 1.2 \times 10^4 (k_{\text{air}})$
	<i>coded:</i> <code>1/diff_sub</code>	1.29×10^5 ×

Over-weighted by $11\times$ to $1300\times$ depending on where in the interface k is evaluated — and `ThermalCond` interpolates between the two, so *any* reasonable choice leaves this term negligible. That is 13.1 of the 17.8 percentage points, and it is wrong regardless of how the asymptotics behave.

The vapour term (the remaining 4.7 points) is structurally *correct* but still is not delivered. Most likely — **conjecture, not established** — because vapour diffusion is one-sided through `phi_aef`, the case that requires an anti-trapping current the solver does not have; the symmetric-model correction simply does not apply there.

4.1 Candidate fixes

τ_{sub}	$\beta_{\text{bare}}/\beta_0$	predicted ratio
as coded	1.2165	0.8220
thermal term fixed (mean k)	1.0572	0.9459
both a_2 terms dropped	1.0000	1.0000
measured	—	0.8259 (L) / 0.8109 (R)

5. ϵ is not W_{KP} — but a_1, a_2 already absorb it

This model’s well gives the equilibrium profile $\phi = \frac{1}{2}[1 + \tanh(x/2\epsilon)]$; Karma’s gives $\frac{1}{2}[1 + \tanh(x/\sqrt{2}W)]$. Hence

$$\boxed{W_{\text{KP}} = \sqrt{2}\epsilon} \quad (5)$$

Mapping the $[0, 1]$ equation onto Karma’s $[-1, 1]$ form gives $\tau_{\text{KP}} = 2\tau$, $W = \sqrt{2}\epsilon$, $\lambda_{\text{KP}} = \lambda/4$, and therefore

$$d_0 = a_1^{\text{KP}} \frac{W}{\lambda_{\text{KP}}} = 0.883883 \cdot \sqrt{2} \cdot 4 \cdot \frac{\epsilon}{\lambda} = 5.0000 \frac{\epsilon}{\lambda} \quad \text{code: } a_1 = 5.0 \quad (0.000\%) \quad (6)$$

$$C_{\text{corr}} = a_1^{\text{KP}} a_2^{\text{KP}} \frac{W}{D} = 0.783375 \frac{\epsilon}{D} \quad \text{code: } a_1 a_2 = 0.7905 \quad (0.91\%) \quad (7)$$

So the $\sqrt{2}$ is real and already accounted for. Independently confirmed: the free fit recovers d_0 to 0.05%, which is a direct measurement of $a_1 = 5$.

6. The bumps in v_n are the reading, not the ice

The original periodic ripple satisfied $|v_n| T_{\text{ripple}} = 6.0 \times 10^{-7}$ m on every moving front (0.90–1.06 $\times dx$ over twelve fronts): one cycle per control-point spacing traversed. Each front has its own

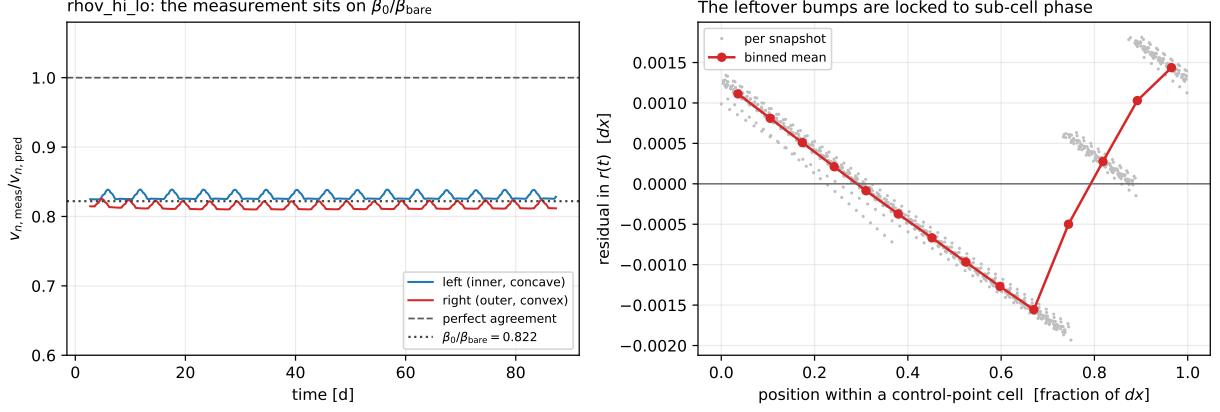


Figure 2: **Left:** the measured ratio sits on $\beta_0/\beta_{\text{bare}}$ for the whole run. **Right:** the leftover wobble is a deterministic function of where the interface sits within a control-point cell — extraction bias, not ice.

period simply because $T = dx/|v_n|$ and the fronts move at different speeds. The wobble in $r(t)$ falls $15\times$ as the reading improves, which is what rules out lattice pinning (real stick-slip would be identical in all four):

wobble in $r(t)$, units of dx	2-pt linear	tanh fit
control net (p=2 B-spline coefficients)	0.0026	0.0009
true NURBS, $4\times$	0.0012	0.0002

Two causes compound: the `.vts` carries *control coefficients*, not field values ($\sim 10^{-9}$ m of position error that drifts in phase as the interface crosses the grid), and a two-sample $\phi = 0.5$ crossing is biased by where those samples fall. The default is now `--interface tanh`; the residual v_n wobble is 0.5 % and, as the figure shows, still locked to sub-cell phase.

Is it biasing the answer? No. Three independent estimators agree on the median:

estimator	median vs. reference	wobble
position difference + sliding fit (current default)	—	0.49 %
level set $-\phi_t/\phi_x$, ϕ_x from <code>np.gradient</code>	+4.5 %	0.13 %
level set, $\phi_x = 1/(4\epsilon)$ analytic	+1.0 %	0.95 %

The +4.5 % is fully explained: a centred difference underestimates $|\nabla\phi|$ of a tanh on this grid by $1 - \tanh(h)/h = 3.94\%$ with $h = dx/2\epsilon$. Correcting for it brings that estimator within 0.2–0.4 % of the default *and* keeps the 0.13 % wobble — available as an upgrade if ever needed.

6.1 In-solver or post-processing?

Post-processing, for now. The extraction is at 0.5 % against an effect of 18 %, and the medians are confirmed by independent estimators. An in-solver $v_n = -\phi_t/|\nabla\phi|$ at quadrature points would use the true NURBS field and the integrator’s own ϕ_t , eliminating both grid-phase bias and snapshot differencing — worth doing if (a) a τ_{sub} fix must be validated to $\sim 1\%$ across a sweep, or (b) the check is wanted on geometries with no convenient centreline. Not needed to settle the present question.

7. The test: a β_{sub0} ladder

Four runs, identical but for β_{sub0} . Because the a_2 inflation is *additive* and β_{sub0} is not, the predicted shortfall is a strong function of β_{sub0} . `-t_final` and `-dtmax` are set per arm so all four run the same 3406 steps and move the menisci the same ~ 2 control-point spacings — equal cost, equal precision.

$\times$	<code>-beta_sub0</code>	predicted ratio	τ_{sub} [s]	<code>-dtmax</code>	<code>-t_final</code>
0.10	5.9216e4	0.3160	135.8	1.485e2	5.0577e5 (5.9 d)
0.25	1.4804e5	0.5359	200.2	2.188e2	7.4548e5 (8.6 d)
1.00	5.9216e5	0.8220	522.1	5.707e2	1.9440e6 (22.5 d)
4.00	2.3686e6	0.9487	1809.8	1.978e3	6.7381e6 (78.0 d)

Reading the outcome (meas / pred from each run’s summary):

- 0.316/0.536/0.822/0.949 — the a_2 terms are never subtracted. Fix τ_{sub} .
- 1.00 at all four — they *do* operate; the post-processing is wrong somewhere.
- 0.82 at all four — neither story; the deficit is a constant factor from something else.

To run it:

```
./scripts/HPC/submit_batch.sh --tag beta_ladder \
  --tests-file scripts/HPC/tests_beta_ladder.txt
```

then, after `rsync`’ing the batch down, `bash <batch>/run_batch_postprocess.sh`.

8. Open items, in priority order

1. Run the ladder. Everything below is conditional on its outcome.
2. Fix the thermal a_2 term (Clausius–Clapeyron factor). Unambiguous; 13.1 of 17.8 points.
3. Decide the vapour a_2 term: drop it, or add an anti-trapping current so the correction it assumes is actually valid. The latter is a real modelling project.
4. Document the ξ_v mass-conservation bound. Not a bug; a stated approximation.
5. Optional: in-solver v_n diagnostic, if the GT check is wanted on general geometries.